KASHMIR COLLEGE OF ENGINEERING & TECHNOLOGY

Approved by AICTE New Delhi | Affiliated with the University of Kashmir
Campus: Ranbirgarh Narbal, Srinagar - 190017, Jammu & Kashmir | Web: https://kcet.edu.in

B.Tech in Information Technology - Complete Syllabus

Approved AICTE 4-Year Scheme & Detailed Syllabus Booklet | Category: Syllabus
Curriculum Model: AICTE Model Curriculum / University of Kashmir Ordinances

Semester 3

* IT-301: Discrete Mathematical Structures (4 Credits)
* IT-302: Data Structures & Algorithms (4 Credits)
* IT-303: Digital Electronics & Logic Design (3 Credits)
* IT-304: Object-Oriented Methodology using Java (3 Credits)
* IT-305: Principles of Communication Systems (3 Credits)
* IT-306P: Data Structures Lab (2 Credits)
* IT-307P: Java Programming Lab (2 Credits)

Semester 4

* IT-401: Computer Organization & Microprocessors (4 Credits)
* IT-402: Database Management Systems (4 Credits)
* IT-403: Design & Analysis of Algorithms (4 Credits)
* IT-404: Operating Systems Concepts (3 Credits)
* IT-405: Software Engineering Principles (3 Credits)
* IT-406P: DBMS & SQL Lab (2 Credits)
* IT-407P: Operating Systems & Shell Programming Lab (2 Credits)

Semester 5

* IT-501: Computer Communication Networks (4 Credits)
* IT-502: Formal Languages & Automata Theory (3 Credits)
* IT-503: Web Development Frameworks (React & Node.js) (3 Credits)
* IT-504: Information Storage & Management (3 Credits)
* IT-PE1: Professional Elective I: Cyber Forensics & Investigation (3 Credits)
* IT-505P: Web Engineering & Frameworks Lab (2 Credits)
* IT-506P: Computer Networks & Protocol Lab (2 Credits)

Semester 6

* IT-601: Information & Cryptographic Security (4 Credits)
* IT-602: Software Project Management & Agile (3 Credits)
* IT-603: Mobile Computing & Wireless Networks (3 Credits)
* IT-604: Data Science & Analytics with Python (4 Credits)
* IT-PE2: Professional Elective II: Enterprise Resource Planning (ERP) (3 Credits)
* IT-605P: Mobile Application Development Lab (2 Credits)
* IT-606P: Data Science & Analytics Lab (2 Credits)

Semester 7

* IT-701: Cloud Computing & Virtualized Services (4 Credits)
* IT-702: Big Data Technologies & Hadoop Ecosystem (3 Credits)
* IT-PE3: Professional Elective III: DevOps & Continuous Integration (3 Credits)
* OE-701: Open Elective I: IT Laws & Patenting (3 Credits)
* IT-703P: Capstone Project Stage-I (4 Credits)
* IT-704I: Industrial Summer Internship Evaluation (2 Credits)

Semester 8

* IT-801: Major Capstone Project Stage-II (Working IT Solution) (10 Credits)
* IT-PE4: Professional Elective IV: Artificial Intelligence & Deep Learning (3 Credits)
* OE-801: Open Elective II: Human-Computer Interaction (HCI) (3 Credits)
* IT-802V: Comprehensive Technical Viva-Voce (2 Credits)